Navleen Singh

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BASc Mechatronic Systems Engineering · Simon Fraser University

TECHNICAL SKILLS

Software: SolidWorks, FluidSIM, MATLAB, TIA Portal, Arduino IDE

Programming: Embedded C, Python, C++, SDL2, OpenCV

Electrical / Hardware: Electropneumatics, CAN bus, OBD-II, Festo Didactic systems, SPI, multimeter, oscilloscope

Certifications: Siemens Mechatronic Systems Certification Program (SMSCP) — Assistant · May 2026 · Cert. No. 1-41-0241

EDUCATION

Simon Fraser University Sept 2025 – Present

Bachelor of Applied Science, Mechatronic Systems Engineering | Expected: Apr 2028

University of the Fraser Valley June 2023 – June 2025

Diploma in Computer Science

ENGINEERING PROJECTS

Autonomous Robotics: Computer Vision, Inverse Kinematics and Machine Learning Sept 2025

SFU, Mechatronic Systems Engineering

- Developed an autonomous robot on Raspberry Pi with PID-controlled line following using OpenCV, ultrasonic obstacle detection, and a pick-and-place arm controller using inverse kinematics for real-time closed-loop control.
- Integrated YOLOX object detection with C++ inference bridged to a Python controller over IPC; benchmarked model sizes for embedded deployment and implemented automated shape classification using computer vision techniques.

CAN Bus Vehicle Data Logger Feb 2026

Personal Project

- Designed a hardware data logger using an MCP2515 CAN transceiver over SPI with embedded C firmware; captures OBD-II parameters including engine load, boost pressure, exhaust temperature, and injection timing at 500 kbps with timestamped data logged to SD card in CSV format.

Vehicle Audio System: Electrical Integration and Fault Diagnosis May 2026

Personal Project

- Designed a 320W 4-channel amplifier installation including power distribution, wire sizing, fuse selection, chassis ground verification, and CAN bus remote turn-on integration; diagnosed inconsistent channel output through systematic electrical isolation and identified a defective RCA cable as the root cause.

Automotive Charging System Diagnosis and Repair May 2026

Personal Project

- Applied structured electrical troubleshooting to diagnose alternator failure: performed voltage comparisons, fuse continuity testing, and direct B+ terminal measurement under load; confirmed repair with post-installation voltage validation.

Siemens PLC Station Programming May 2026

SMS103 Siemens Mechatronic Systems Certification | SMSCP Assistant

- Programmed a Siemens S7 automation station in TIA Portal including symbol tables, ladder logic for sequential control and interlocks, and function block diagrams for pneumatic sequencing; commissioned on Festo industrial hardware with full fault detection validation.

Electropneumatic Control Systems Mar 2026

SMS103 Siemens Mechatronic Systems Certification | SMSCP Assistant

- Designed and validated direct and indirect electropneumatic circuits using double-solenoid DCVs and relay logic in FluidSIM simulation before physical implementation on Festo hardware; documented a complete BOM and I/O assignment table for PLC integration.

Pneumatic Membrane Braille Tablet Jan 2026

SFU, Engineering Design Project

- Led a 5-person team through the full mechanical design cycle in SolidWorks for an affordable pneumatic braille display; conducted FMEA on the actuation system and demonstrated feasibility at a sub-$50 CAD unit cost through comparative cost analysis.

Snake Game: Modular C and SDL2 Software Project Jan 2026

SFU, Group Software Engineering Project

- Built a modular Snake game in C with SDL2 using a singly linked list, dynamic memory management with zero heap leaks, a 4-state game state machine, and clean separation across logic, physics, and rendering modules in a 4-person codebase.

WORK EXPERIENCE

Soccer Referee | VMSL, FVSL, BMSL, KSL, BCCSL, BCPL Sept 2018 – Present

Multiple Leagues, Men's and Youth | Surrey, BC

- Officiated across 6 competitive leagues over 7 years managing player conduct, applying rule judgements under pressure, and maintaining consistent performance standards across men's and youth divisions.
- Selected for the 2026 BC Soccer Referee Provincial Program, an invitation-only provincial development camp recognizing high-performing referees.

LANGUAGES

Italian (Native) · Punjabi (Fluent) · English (Full Professional Proficiency) · Hindi (Conversational) · French (Conversational)

INTERESTS

Automotive ECU calibration and powertrain systems including Bosch EDC17 fueling and boost mapping · Vehicle network architecture using CAN and LIN protocols · Embedded firmware development and microcontroller interfacing · Industrial automation with Siemens PLC ecosystems and SCADA integration · ADAS and autonomous systems including sensor fusion and computer vision pipelines